

Project 3-1: Letter Grade Converter

Create a program that converts number grades to letter grades.

Console

```
Letter Grade Converter

Enter numerical grade: 90
Letter grade: A

Continue? (y/n): y

Enter numerical grade: 88
Letter grade: A

Continue? (y/n): y

Enter numerical grade: 80
Letter grade: B

Continue? (y/n): y

Enter numerical grade: 67
Letter grade: C

Continue? (y/n): y

Enter numerical grade: 59
Letter grade: F

Continue? (y/n): n

Bye!
```

Specifications

- The grading criteria is as follows:

A	88-100
B	80-87
C	67-79
D	60-66
F	<60
- Assume that the user will enter valid integers for the grades.
- The program should continue only if the user enters “y” or “Y” to continue.

Project 3-2: Tip Calculator

Create a program that calculates three options for an appropriate tip to leave after a meal at a restaurant.

Console

```
Tip Calculator

Cost of meal: 52.31

15%
Tip amount:    7.85
Total amount:  60.16

20%
Tip amount:    10.46
Total amount:  62.77

25%
Tip amount:    13.08
Total amount:  65.39
```

Specifications

- The program should calculate and display the cost of tipping at 15%, 20%, or 25%.
- Assume the user will enter valid data.
- The program should round results to a maximum of two decimal places.

Project 3-3: Change Calculator

Create a program that calculates the coins needed to make change for the specified number of cents.

Console

```
Change Calculator

Enter number of cents (0-99): 99

Quarters: 3
Dimes:    2
Nickels:  0
Pennies:  4

Continue? (y/n): y

Enter number of cents (0-99): 55

Quarters: 2
Dimes:    0
Nickels:  1
Pennies:  0

Continue? (y/n): n

Bye!
```

Specifications

- The program should display the minimum number of quarters, dimes, nickels, and pennies that one needs to make up the specified number of cents.
- Assume that the user will enter a valid integer for the number of cents.
- The program should continue only if the user enters “y” or “Y” to continue.

Project 3-5: Table of Powers

Create a program that displays a table of squares and cubes for the specified range of numbers.

Console

```
Table of Powers

Start number: 90
Stop number: 100

Number      Squared      Cubed
=====
90 8100    729000
91 8281    753571
92 8464    778688
93 8649    804357
94 8836    830584
95 9025    857375
96 9216    884736
97 9409    912673
98 9604    941192
99 9801    970299
100 10000  1000000
```

Specifications

- The formulas for calculating squares and cubes are:
`square = x ** 2`
`cube = x ** 3`
- Use tabs to align the columns.
- Assume that the user will enter valid integers.
- Make sure the user enters a start integer that's less than the stop integer. If the user enters a start integer that's greater than the stop integer, display an error message and give the user a chance to enter the integers again.